Towards a Cartography of Datacentering

A Multimodal Research Agenda for Innovation Studies

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The Empirical Moment

Metric	Value
Hyperscaler capex 2026	**$750B**
DC electricity 2025	485 TWh (+17%)
AI-specific growth	+50% YoY
Hyperscaler share 2031	67% of all DC

CreditSights Feb 2026; IEA Energy & AI April 2026; Synergy Research Group

"The largest concentrated infrastructure build of our lifetimes — happening now."

Not a future scenario.

Not a policy proposal.

An arena in formation.

Why Now? The Contested Arena

- **Stargate:** $500B / ∼7 GW; OpenAI + Oracle + SoftBank
- **PJM electricity:** capacity prices **+833%** in one year
- **Maine:** first US statewide moratorium, April 2026
- **Loudoun Co.:** eliminates by-right DC, March 2025
- **DeepSeek V4-Pro:** re-opens Jevons paradox debate

37,552 DCD articles

2008–2026, headline + intro

1,073 contestation events

Peak contestation: 2025

Trade press = primary event record before academic literature

The Gap in Innovation Studies

RP canon covers:

Platforms (Gawer 2014)
GPTs (Brynjolfsson)
Sectoral systems (Malerba)
Transitions (Geels)

Critical DC studies exists:

Big Data & Society, *Convergence*
Edwards et al. 2025; Monstadt 2025
— but outside RP's canon

Missing:

Integrative framework for the *infrastructural + methodological* turn in AI-era innovation studies

No unified approach to:
- mapping contested arenas
- surfacing implicated actors
- operationalizing multimodal evidence

Theory: Arenas of Development

Jørgensen (2012) — *Research Policy*

Flat ontology. Start from *actor constellations* and their collective engagement — not structural tiers.

Negotiation is the fundamental process.

Allows political, social, environmental spheres simultaneously.

+ Clarke (2005): situational maps

Implicated actors: silenced (communities, visa workers) + constructed but absent (“the consumer”, “future generations”)

The datacentering arena:

Co-constitutive.
Not hierarchically ordered.

Theory: Issue Mapping + CITC

Marres (2015) — *ST&HV*

Issue mapping via digital traces.
Trade press, hyperlinks, hashtags as empirical window into issue formation.

Berente et al. (2019); Miranda et al. (2022)

Computationally Intensive Theory Construction

Sampling → synchronic → diachronic → **lexical framing**

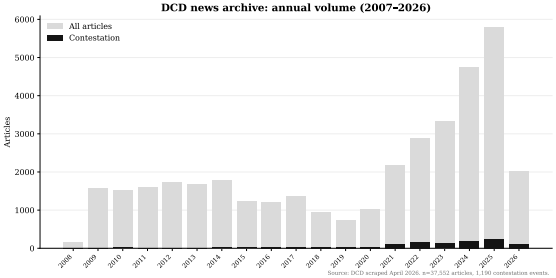
Patterns become contributions *only* once situated in scholarly discourse.

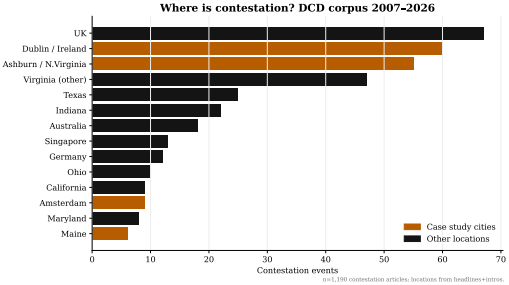
Marres tells us
what to trace

CITC tells us
how to trace it
rigorously enough for RP

Prevents atheoretical pattern-mining.

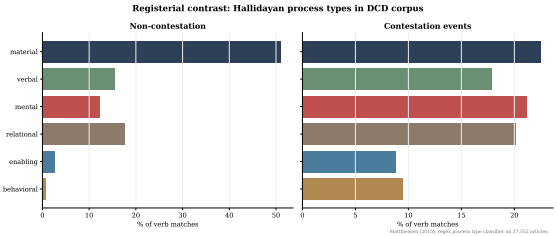
The Arena in Data





37,552 articles , extperiodcentered, 1,073 contestation events , extperiodcentered, peak 2025. **Headlines + intro text only** — full articles via screenshot/OCR pipeline. More coming.

The Register Shift



Matthiessen (2015) registerial cartography — verb process types

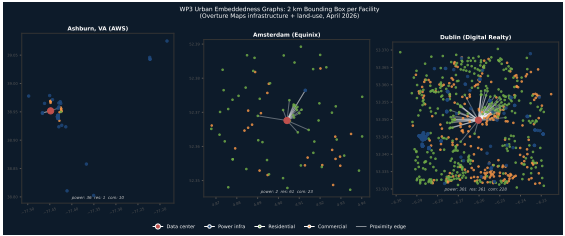
Type	Contest	Non
Behavioral	23.1%	1.6%
Enabling	16.5%	3.1%
Material	33.7%	66.8%

Doing → governing + contesting

The governance gap visible in the verb register.

Headlines + intros only.

City2Graph: Urban Embeddedness



From corpus cities → urban graphs

GLiNER NER extracts city names
→ city2graph pulls Overture Maps
→ graph: DC node ↔ power, residential, commercial

	Res. zones	Power
Ashburn	1	36
Amsterdam	61	2
Dublin	361	6

Spatial embeddedness determines governance counterparty availability.

Three Research Agendas

Agenda	Central question	Cases
Regional Innovation Systems	How do hyperscale clusters reshape local productivity, labour, energy, property?	Loudoun (US), Luleå (SE), Dublin (IE), Hyderabad (IN)
Sectoral Transformation	How does hyperscaler vertical integration reshape sectoral systems and startup access?	AWS, Google, Meta, Microsoft platform dependency
Policy Sovereignty & Energy	What policy mix sustains sovereign compute without fossil lock-in or geopolitical dependency?	UK AI Growth Zones, EU AI Factories, France, Saudi HUMAIN, UAE Stargate

Each agenda operationalized via the CITC cycle (Berente et al. 2019): sampling → synchronic → diachronic → lexical framing.

Building the Cartography Together

Why open is theoretical necessity

CAS-like phenomena are too distributed, too opaque for any single team.

What we share

- Facility index (CC BY 4.0)
- NLP / CV / network pipelines (MIT)
- DCD contestation corpus
- Register + NER analysis scripts
- `github.com/babajideowoyele/datacentering-cartography`

Concrete asks

- Local planning records + disputes
- Non-English corpus (NL, DE, FR, IE)
- Company registry / REIT data access
- Replication in other jurisdictions
- Methodological critique

*"What would innovation studies look like if we treated data infrastructure as **foundational** rather than peripheral?"*

Q: Isn’t this STS dressed up as innovation studies?

Live: “Fair concern — I’m using Jørgensen and Marres specifically because they bridge to RP and *ST&HV*, not importing critical theory wholesale.”

The methodological spine — CITC (Berente, Seidel & Safadi 2019; Miranda et al. 2022) — comes from *MIS Quarterly* and *ISR*, not from STS. The agenda is testable, not hermeneutic.

Q: Where's the empirical contribution?

Live: "The ECR paper is the agenda. The corpus analysis is the first empirical instantiation."

From 37,552 DCD articles:

Analysis	Status	Finding
EDA geography	Complete	US/Europe split; peak 2025
Register analysis	Complete	Behavioral +21.5%, enabling +13.4%
BERTopic topics	Running	Full corpus topic clusters
city2graph	Seed run	3 cities; scales via GLiNER NER

Analysis on headlines + intros. Full article text: screenshot/OCR pipeline coming.

Q: Why Loudoun, Luleå, Dublin, Hyderabad?

Variation on three dimensions:

Case	Regulatory	Energy	Trajectory
Loudoun	Common law / NIMBY	Gas + solar	Mature US cluster
Luleå	Swedish planning	98% hydro	Renewable-led
Dublin	Irish planning	Mixed	FDI platform
Hyderabad	Indian regulatory	Expanding	Emerging hub

Q: What are the register verb families?

Following Matthiessen (2015) Hallidayan process types — applied to DCD headlines + intros:

Process type	Field of activity	Top verbs in corpus
Material	Doing / building	secures, builds, launches, withdraws, rejects
Verbal	Reporting	says, announces, claims, warns, demands
Mental	Exploring / sharing	opposes, concerns, challenges, supports
Relational	Expounding	is, becomes, remains, includes
Enabling	Enabling	bans, permits, rezones, mandates, regulates
Behavioral	Contesting	protests, campaigns, sues, appeals, petitions

Contestation = shift from material/relational to behavioral/enabling.

Q: Why CITC and not CSS more broadly?

CITC's lexical-framing requirement is what makes the output theoretical, not just descriptive.

That's the discipline that protects against the most common reviewer objection:

> *"This is interesting, but what's the theoretical contribution?"*

CITC explicitly distinguishes patterns from contributions. The test for RP.